Inverse Problems in Geophysics Part 5: TSVD & Regularization

2. MGPY+MGIN

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Recap problem types/SVD

- over-determined problems: least-squares (LS) solution
- under-determined problems: minimum-norm (MN) solution
- rank determines problem type ($r > M$ or $r > N$)
- singular value decomposition (SVD) as fundamental tool
  - for all cases: the pseudo-inverse (LS & MN special cases)
  - data/model resolution determined by data/model eigenvectors

Inverse problem types - classification scheme

$\mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n} \Rightarrow \mathbf{G} \in \Re^{\mathbf{N} \times \mathbf{M}}$

rows = data (N), columns = model parameter (M), rank r determines type of problem

Even-determined (LS=MN)

$M = N = r$, $\mathbf{R}^M$=$\mathbf{I}$, $\mathbf{R}^D$=$\mathbf{I}$

Over-determined (LS)

$N > r = M$, $\mathbf{R}^M$=$\mathbf{I}$, $\mathbf{R}^D \neq \mathbf{I}$

Under-determined (MN)

$N = r < M$, $\mathbf{R}^M \neq \mathbf{I}$, $\mathbf{R}^D = \mathbf{I}$

Mixed-determined

$r < N$, $r < M$, $\mathbf{R}^M \neq \mathbf{I}$, $\mathbf{R}^D \neq \mathbf{I}$

SVD

(SVD-based) pseudo-inverse provides a unique solution for all cases

The Singular Value Decomposition

$\mathbf{G}$ consists of model & data eigenvectors, weighted by singular values

$$\mathbf{G} = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \cdot \mathbf{v}_i^T = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^T$$

- eigenvalues in $\mathbf{\Sigma} = \mathrm{diag}(\sigma_i) \in \mathbb{R}^{r \times r}$
- orthonormal data eigenvectors $\mathbf{U}_r \in \mathbb{R}^{N \times r}$ with $\mathbf{U}_r^T \mathbf{U}_r = \mathbf{I}$
- orthonormal model eigenvectors $\mathbf{V}_r \in \mathbb{R}^{M \times r}$ with $\mathbf{V}_r^T \mathbf{V}_r = \mathbf{I}$
- generalized inverse $\mathbf{G}^\dagger = \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{U}_r^T = \sum \frac{\mathbf{u}_i^T \mathbf{d}}{\sigma_i} \mathbf{v}_i$

The problem of small eigenvalues

Generalized inverse $\mathbf{G}^\dagger = \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{U}_r^T$ to invert noisy data $\mathbf{Gm} + \mathbf{n}$

$$\mathbf{G}^\dagger(\mathbf{Gm} + \mathbf{n}) = \mathbf{G}^\dagger\mathbf{Gm} + \mathbf{G}^\dagger\mathbf{n} = \mathbf{R}^M + \sum_{i=1}^{r} \frac{\mathbf{u}_i^T\mathbf{n}}{\sigma_i}\mathbf{v}_i$$

Problem

Small eigenvalues amplify noise in the data!

Truncated singular value (TSVD) method

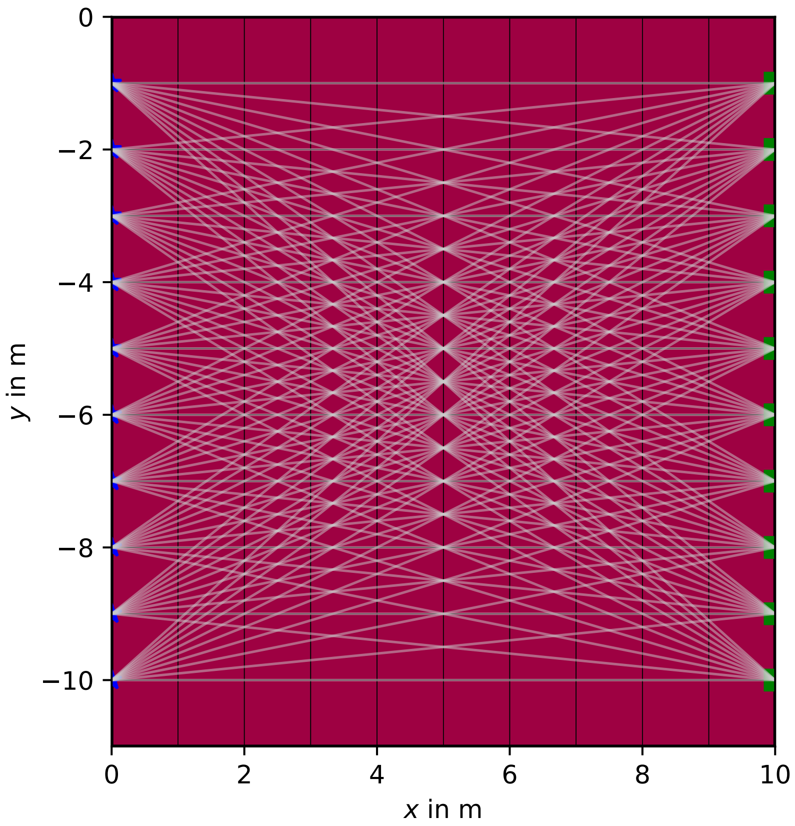
- Look at the singular value spectrum considering noise
- Choose a maximum number p (or `np.linalg.pinv(rcond=)`)

$$\mathbf{G}_p^\dagger = \mathbf{V}_p \mathbf{\Sigma}_p^{-1} \mathbf{U}_p^T$$

- How to choose p? Trade-off between resolution and artifacts.

> **Discrepancy principle (free after Occam)**
>
> Look at data fit and choose p such that the data can be fitted within the error.

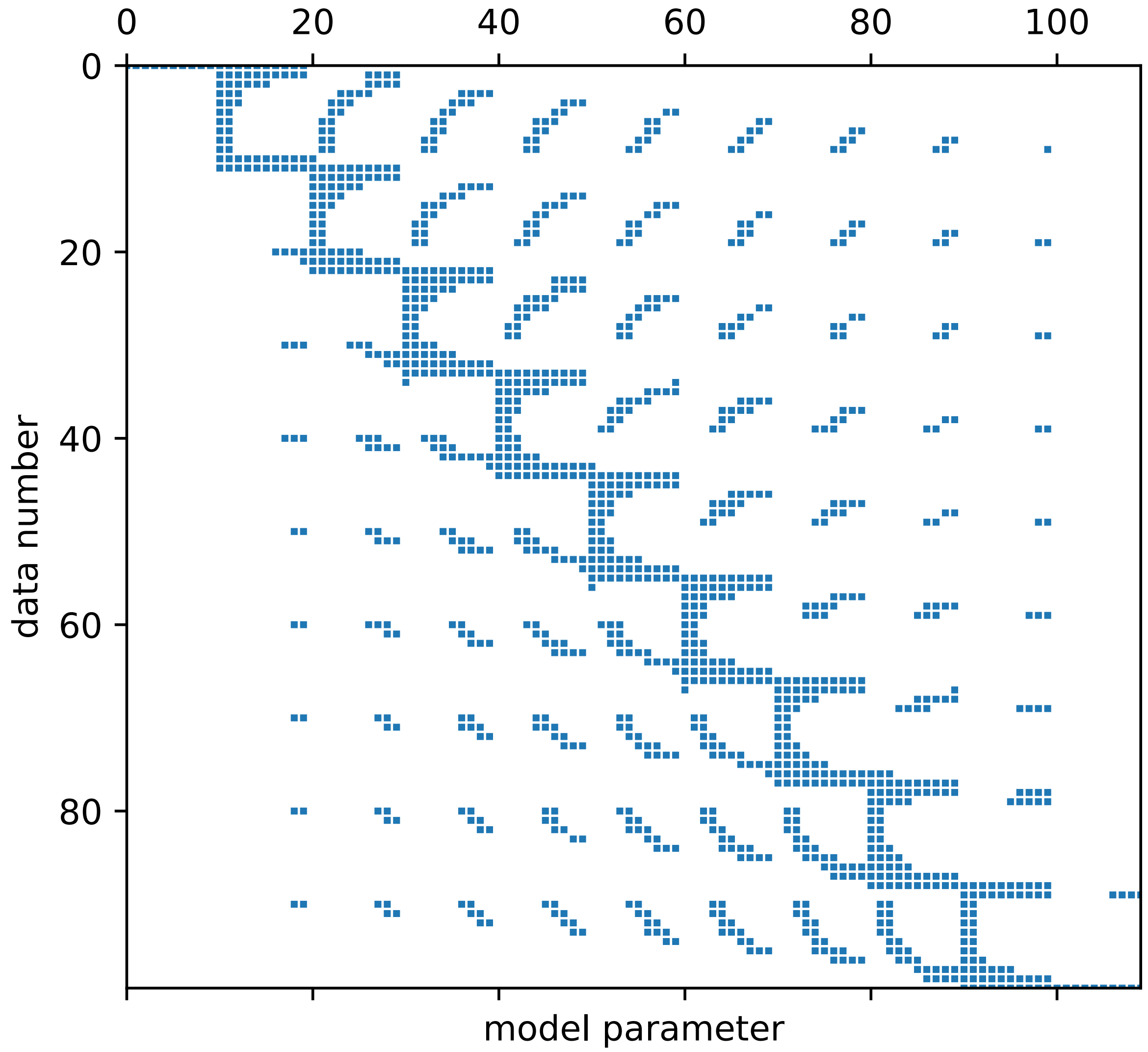
Geophysical example



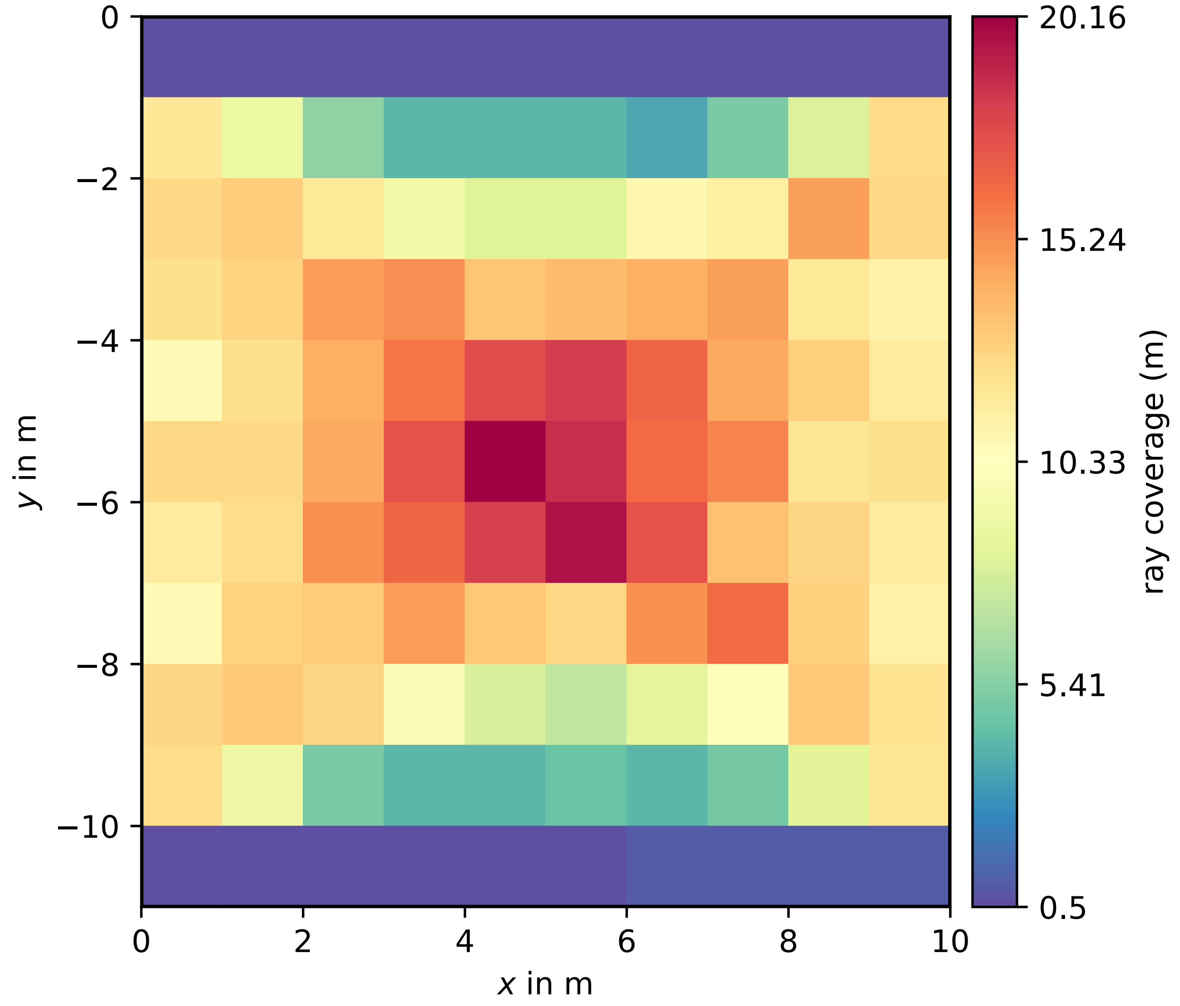
Seismic crosshole tomography

- grid with 1m spacing (11x10 cells)
- two boreholes: shots left, geophones right, fully connected (10x10 data)
- straight ray paths (x-ray, small contrasts)

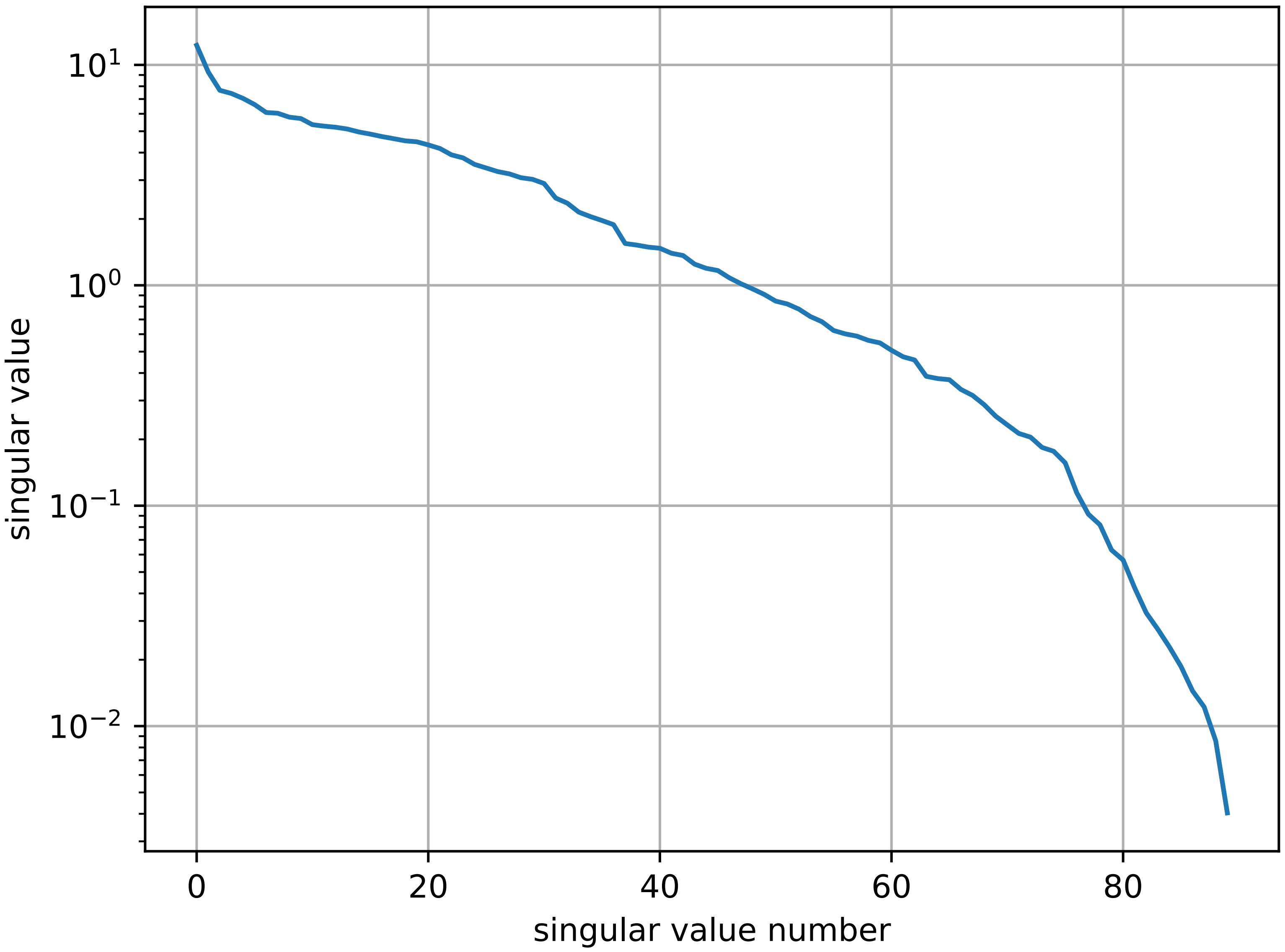
Way matrix (100 $\times$ 110) and coverage



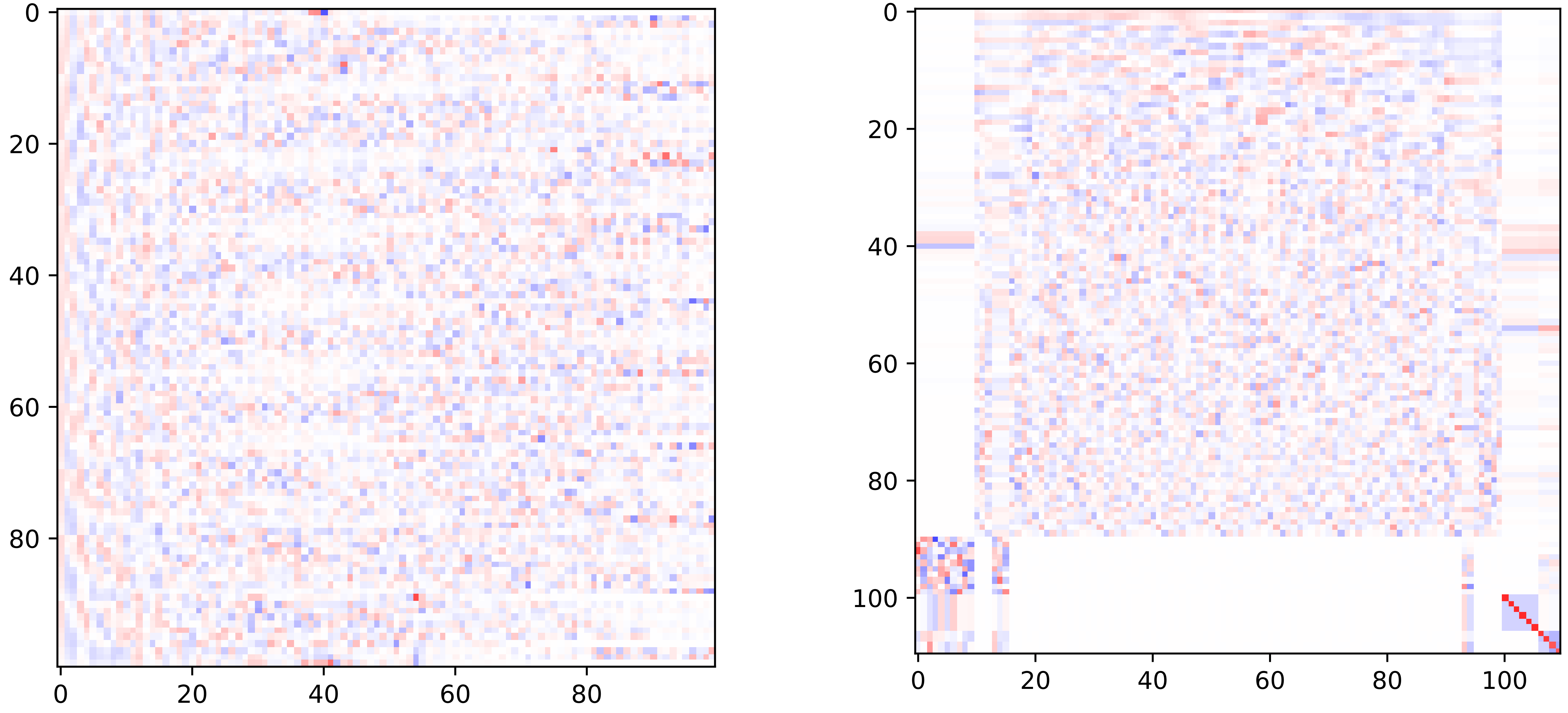
Coverage: data-sum of Jacobian



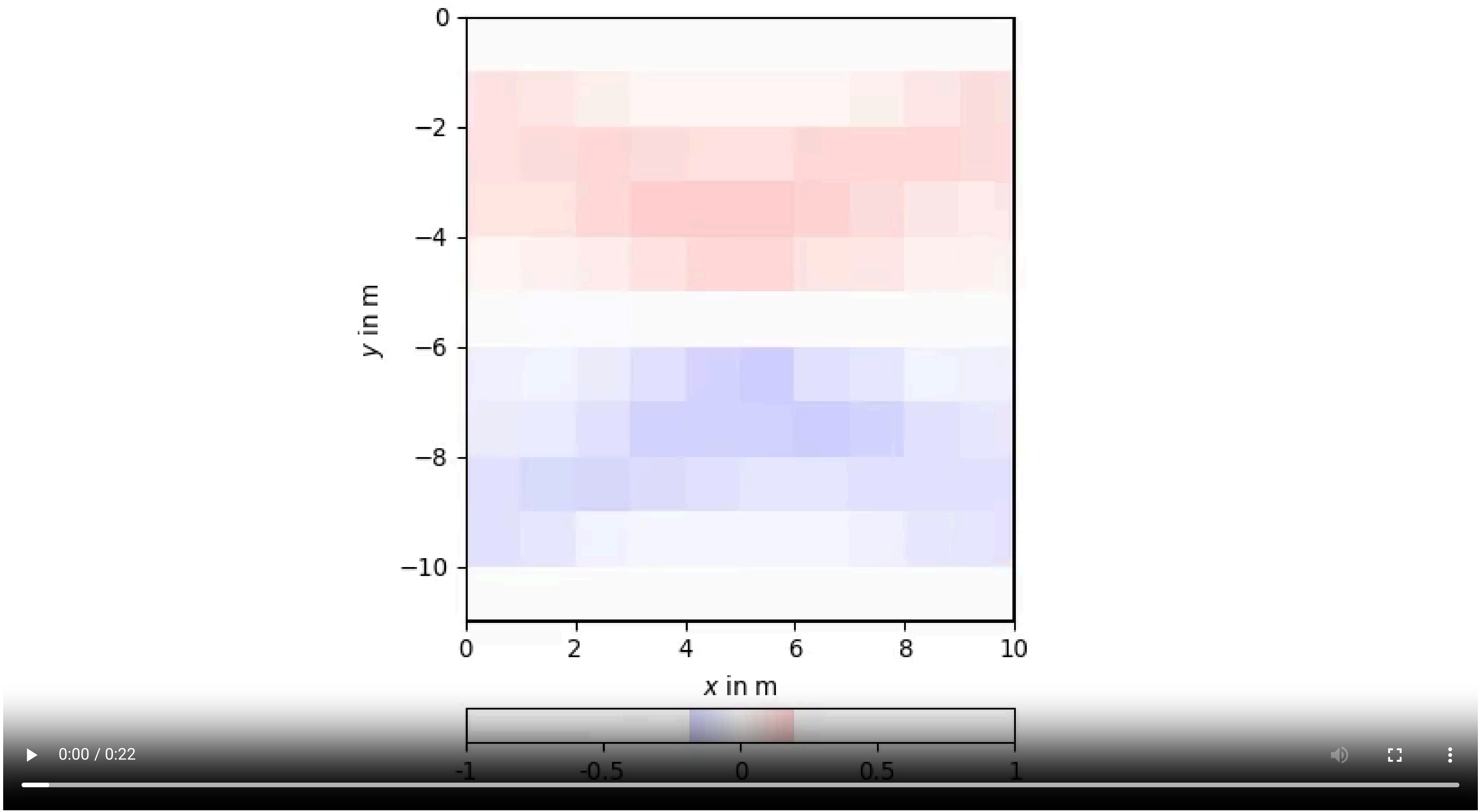
Singular value spectrum



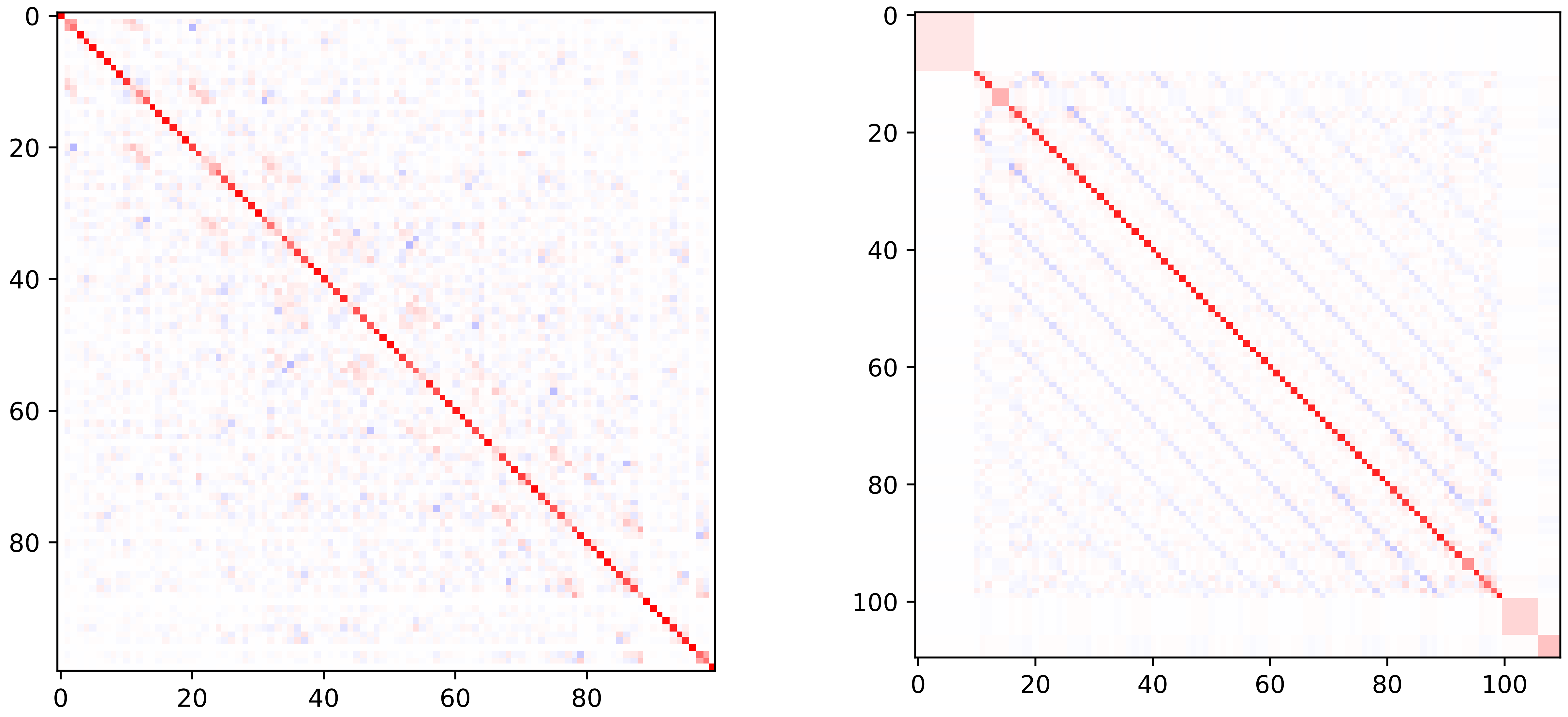
Data (left) and model (right) eigenvectors



Model eigenvectors


0
−2
−4
−6
−8
−10
y in m
0
2
4
6
8
10
x in m
-1
-0.5
0
0.5
1
0:00 / 0:22

Data and model resolution matrix



Regularization

- making under-determined and ill-posed problems unique (regular)
- make the model less sensible to small changes (noise) in data
- adding our assumptions or knowledge (valid ranges, prior data, geostatistical behaviour, interfaces)

Occams razor

Of all possible models, choose the simplest! How to define simple?

Regularization basics: adding equations

$$\mathbf{G} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \tilde{\mathbf{G}}$$

- d_1 measures the mean value of m_1 and m_2
- size of subvector $[m_1, m_2]$ should be small

Minimum norm solution

All model parameters are expected to be (similarly) small

$$\tilde{\mathbf{G}}\mathbf{m} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{m} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ m_1^P \\ m_2^P \\ m_3^P \end{pmatrix}$$

or close to some prior knowledge (d_3, d_4, d_5)

Regularization basics: adding equations

$$\mathbf{G} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tilde{\mathbf{G}}$$

- we only measure the mean value of m_1 & m_2 (as well as m_3)
- difference between m_1 and m_2 should be small

$$\tilde{\mathbf{G}}\mathbf{m} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ 0 \end{pmatrix}$$

Smoothness constraints

Gradient (roughness) between neighboring model parameters

$$\tilde{\mathbf{G}}\mathbf{m} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ 0 \\ 0 \end{pmatrix}$$

Regularization scheme

$\tilde{\mathbf{G}}$ consists of forward part & constraints part

$$\tilde{\mathbf{G}} = \begin{bmatrix} \mathbf{G} \\ \mathbf{C} \end{bmatrix} \quad \text{and} \quad \tilde{\mathbf{d}} = \begin{bmatrix} \mathbf{d} \\ \mathbf{c} \end{bmatrix}$$

now over-determined $\Rightarrow$ least-squares solution $(\tilde{\mathbf{G}}^T\mathbf{G})^{-1}\tilde{\mathbf{G}}^T\tilde{\mathbf{d}}$

$$\tilde{\mathbf{G}}^T\tilde{\mathbf{G}} = \mathbf{G}^T\mathbf{G} + \mathbf{C}^T\mathbf{C} \quad , \quad \tilde{\mathbf{G}}^T\tilde{\mathbf{d}} = \mathbf{G}^T\mathbf{d} + \mathbf{C}^T\mathbf{c}$$

$$\mathbf{m} = (\mathbf{G}^T\mathbf{G} + \mathbf{C}^T\mathbf{C})^{-1}(\mathbf{G}^T\mathbf{d} + \mathbf{C}^T\mathbf{c})$$

Weighting data vs. constraints

$\mathbf{d}$ & $\mathbf{c}$ may have completely different magnitudes & physical units, data/constraints maybe too weak or too strong

$\Rightarrow$ weighting of constraints by regularization parameter λ:

$$\Phi = \|\mathbf{G}\mathbf{m} - \mathbf{d}\|^2 + \lambda\|\mathbf{C}\mathbf{m} - \mathbf{c}\|^2 = \Phi_d + \lambda\Phi_m \to \min$$

λ..regularization strength, Φ_d/Φ_m..data/model objective function

$$\Rightarrow \mathbf{m} = (\mathbf{G}^T\mathbf{G} + \lambda\mathbf{C}^T\mathbf{C})^{-1}(\mathbf{G}^T\mathbf{d} + \lambda\mathbf{C}^T\mathbf{c})$$

Interpretation as constrained minimization (Lagrangian)

$$\Phi = \|\mathbf{Gm} - \mathbf{d}\|^2 + \lambda \|\mathbf{Cm} - \mathbf{c}\|^2 \rightarrow \min$$

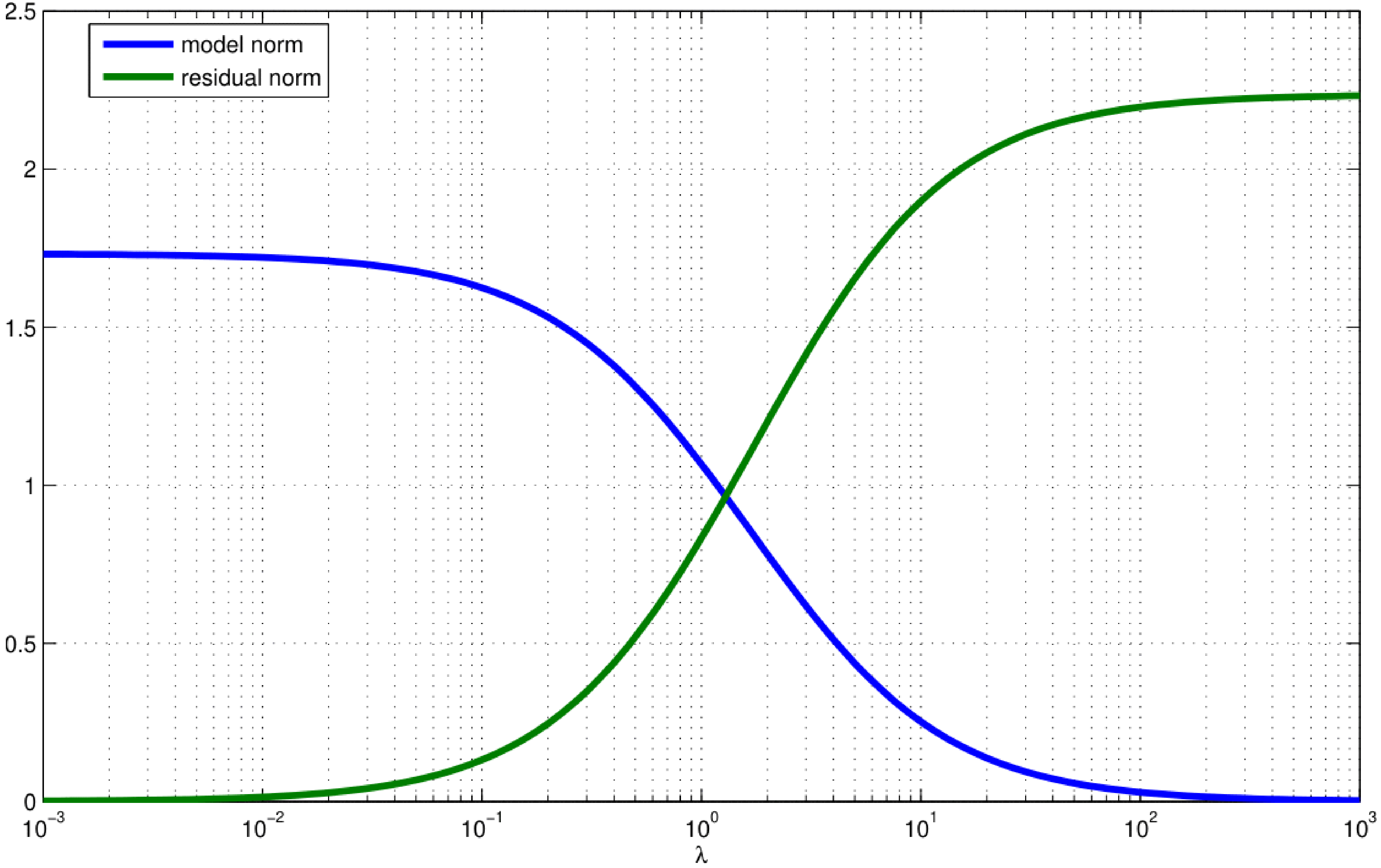
$$\tilde{\Phi} = \|\mathbf{Cm} - \mathbf{c}\|^2 + \mu \|\mathbf{Gm} - \mathbf{d}\|^2 = \Phi_m + \mu \Phi_d \rightarrow \min$$

Minimize regularization term under constraint (Lagrangian method)

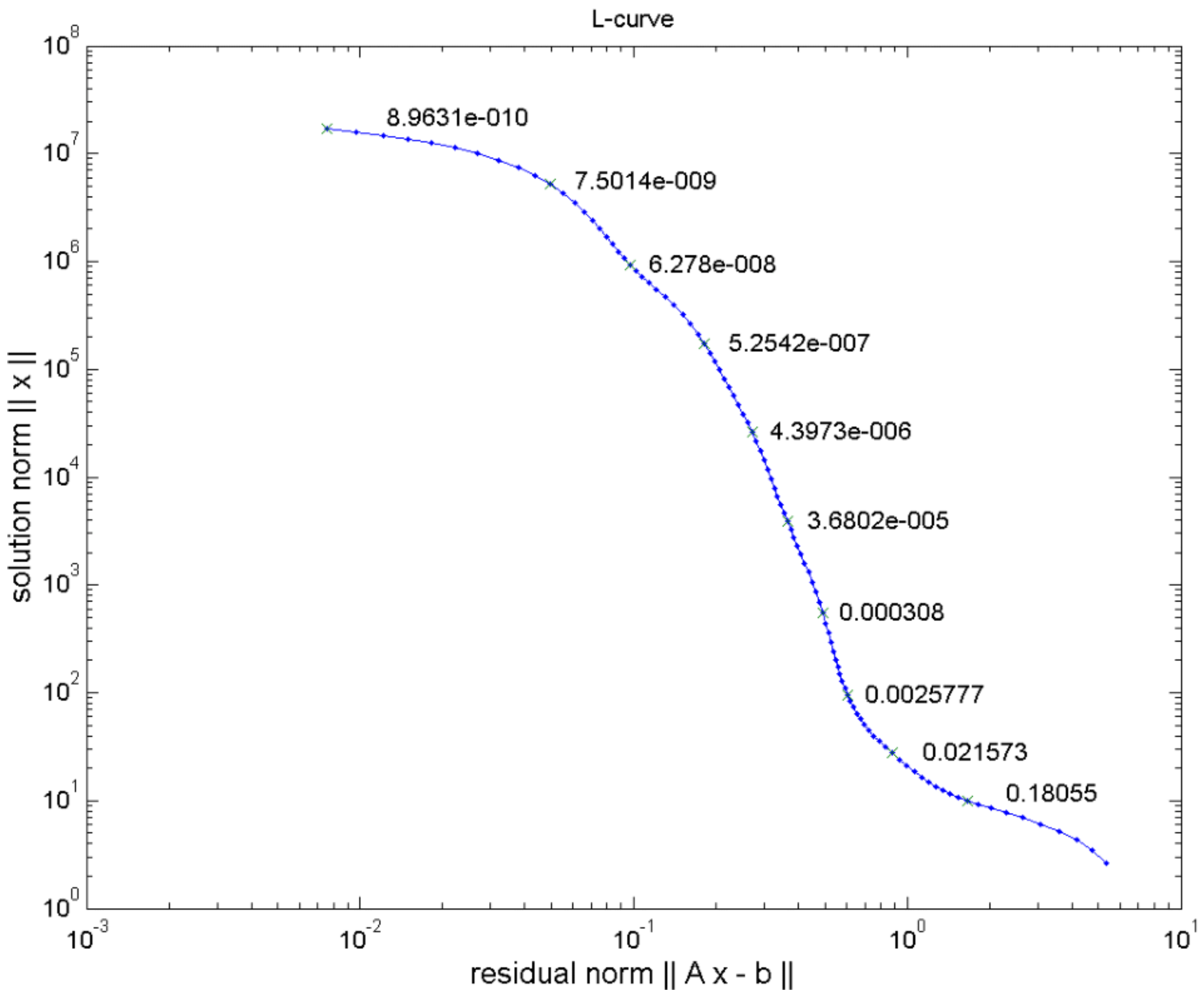
Interpretation with Occam

Minimize the non-simplicity subject to Φ_d reflects noise

Model and data norms



The L-curve



Data vs. model norm for wide range of λ

- low data residual achieved by high norm (oscillating model)
- low model norm cannot fit the data (large misfit)
- optimum somewhere “at the corner” (not always a corner)

Choice of regularization strength

Always have a look at your data fit and model plausibility.

- use different values and look at models (and misfit)
- try to determine the corner of the L-curve (maximum curvature)
- start large λ, decrease & stop when data misfit show no systematics

Discrepancy principle

Choose the highest λ value that is able to fit the data (χ^2=1)!

Damped normal equations and SVD

$$\mathbf{C} = \mathbf{I}, \mathbf{c} = 0 \quad \Rightarrow \quad \mathbf{m} = (\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I})^{-1}\mathbf{G}^T\mathbf{d}$$

$$\mathbf{m} = (\mathbf{V\Sigma U}^T\mathbf{U\Sigma V}^T + \lambda\mathbf{I})^{-1}\mathbf{V\Sigma U}^T\mathbf{d}$$

$$\mathbf{m} = (\mathbf{V}\mathrm{diag}(s_i^2 + \lambda)\mathbf{V}^T + \lambda\mathbf{I})^{-1}\mathbf{V\Sigma U}^T\mathbf{d}$$

$$\mathbf{m} = \sum_i^r \frac{s_i}{s_i^2 + \lambda}\mathbf{u}_i^T\mathbf{d} \cdot v_i^T = \sum_i^r \frac{s_i^2}{s_i^2 + \lambda}\frac{\mathbf{u}_i^T\mathbf{d}}{s_i}v_i^T$$

Small singular values are damped in inversion, large unchanged

Resolution of regularized inverse problems

For $c = 0$ we have $\mathbf{G}^\dagger = (\mathbf{G}^T\mathbf{G} + \lambda\mathbf{C}^T\mathbf{C})^{-1}\mathbf{G}^T$

$$\Rightarrow \mathbf{R}^M = \mathbf{G}^\dagger\mathbf{G} = (\mathbf{G}^T\mathbf{G} + \lambda\mathbf{C}^T\mathbf{C})^{-1}\mathbf{G}^T\mathbf{G}$$

approaches $\mathbf{I}$ for $\lambda \to 0$ and deviates if λ grows

Wrap up

- SVD provides a general tool, BUT:
  - ill-conditioned inverse problems (SV spectrum) tend to amplify noise
  - truncated SVD is a method so suppress this
- regularization can (also) be done to make solution unique
- different strategies: smoothness, norm, …
- choice of regularization strength λ is vital

Application

Damped least squares and SVD

$$\mathbf{m} = (\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I})^{-1}\mathbf{G}^T\mathbf{d}$$

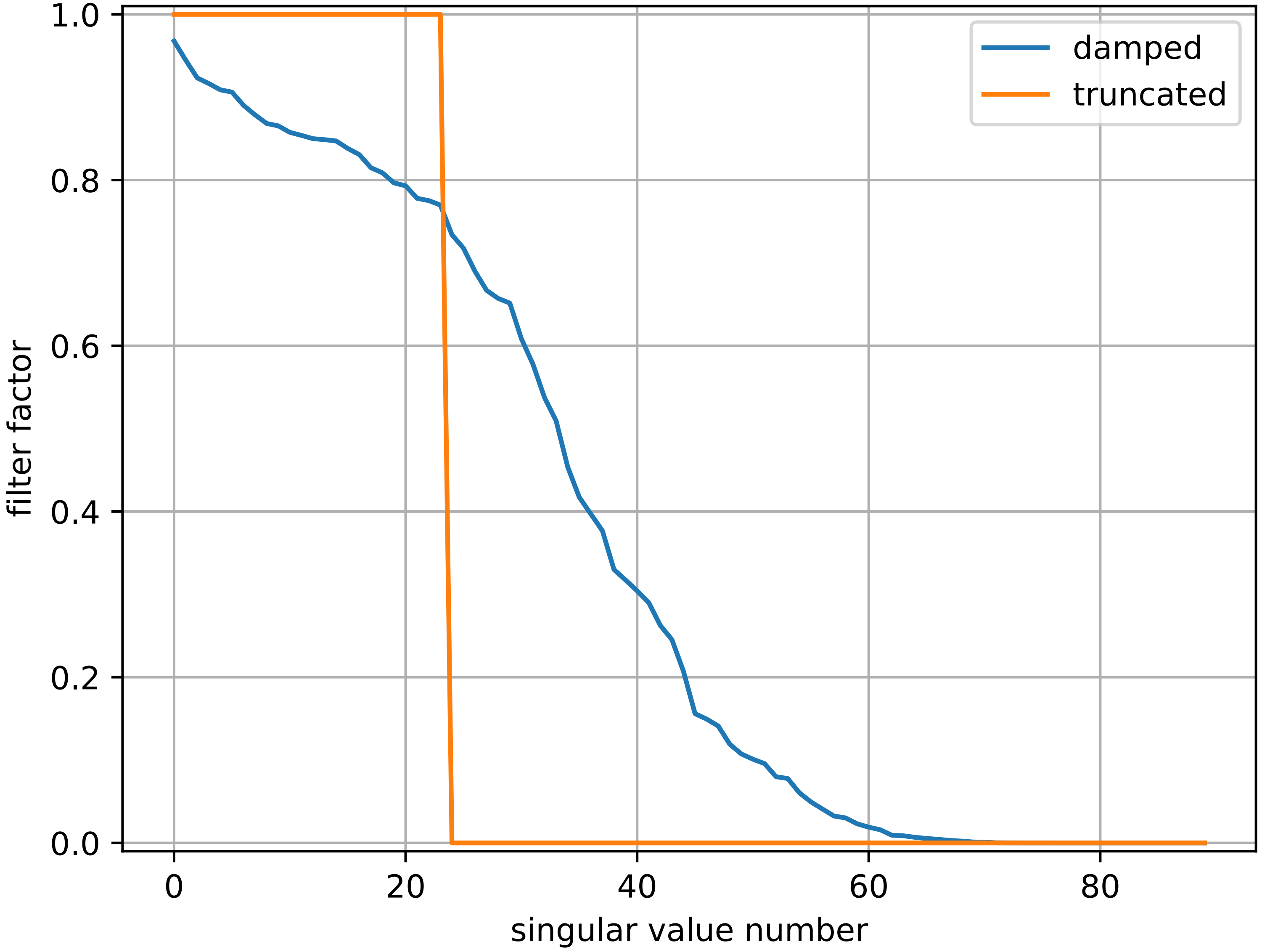
$$\mathbf{m} = (\mathbf{V\Sigma U}^T\mathbf{U\Sigma V}^T + \lambda\mathbf{I})^{-1}\mathbf{V\Sigma U}^T\mathbf{d}$$

$$\mathbf{m} = (\mathbf{V}\mathrm{diag}(s_i^2 + \lambda)\mathbf{V}^T)^{-1}\mathbf{V\Sigma U}^T\mathbf{d}$$

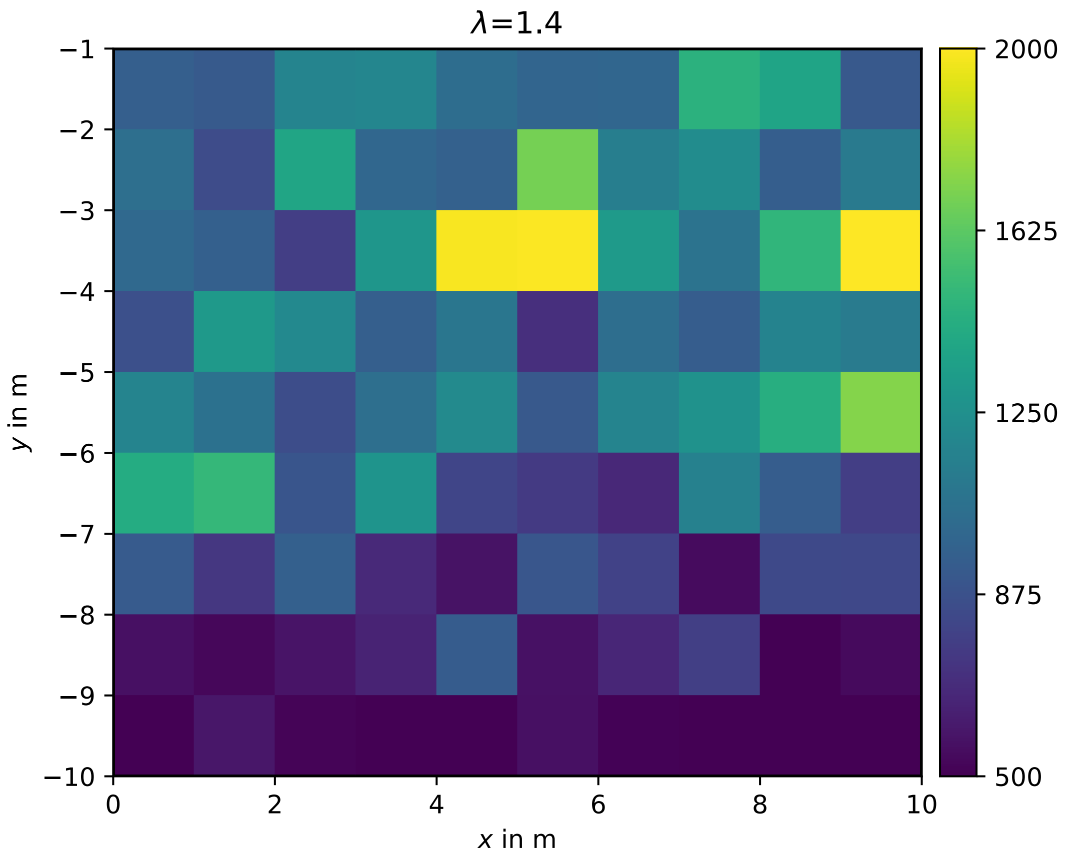
$$\mathbf{m} = \sum_i^r \frac{s_i}{s_i^2 + \lambda}\mathbf{u}_i^T\mathbf{d}\cdot\mathbf{v}_i = \sum_i^r \frac{s_i^2}{s_i^2 + \lambda}\frac{\mathbf{u}_i^T\mathbf{d}}{s_i}\mathbf{v}_i = \sum_i^r f_i\frac{\mathbf{u}_i^T\mathbf{d}}{s_i}\mathbf{v}_i$$

Small singular values are damped by filter factors $f_i = s_i^2/(s_i^2 + \lambda)$

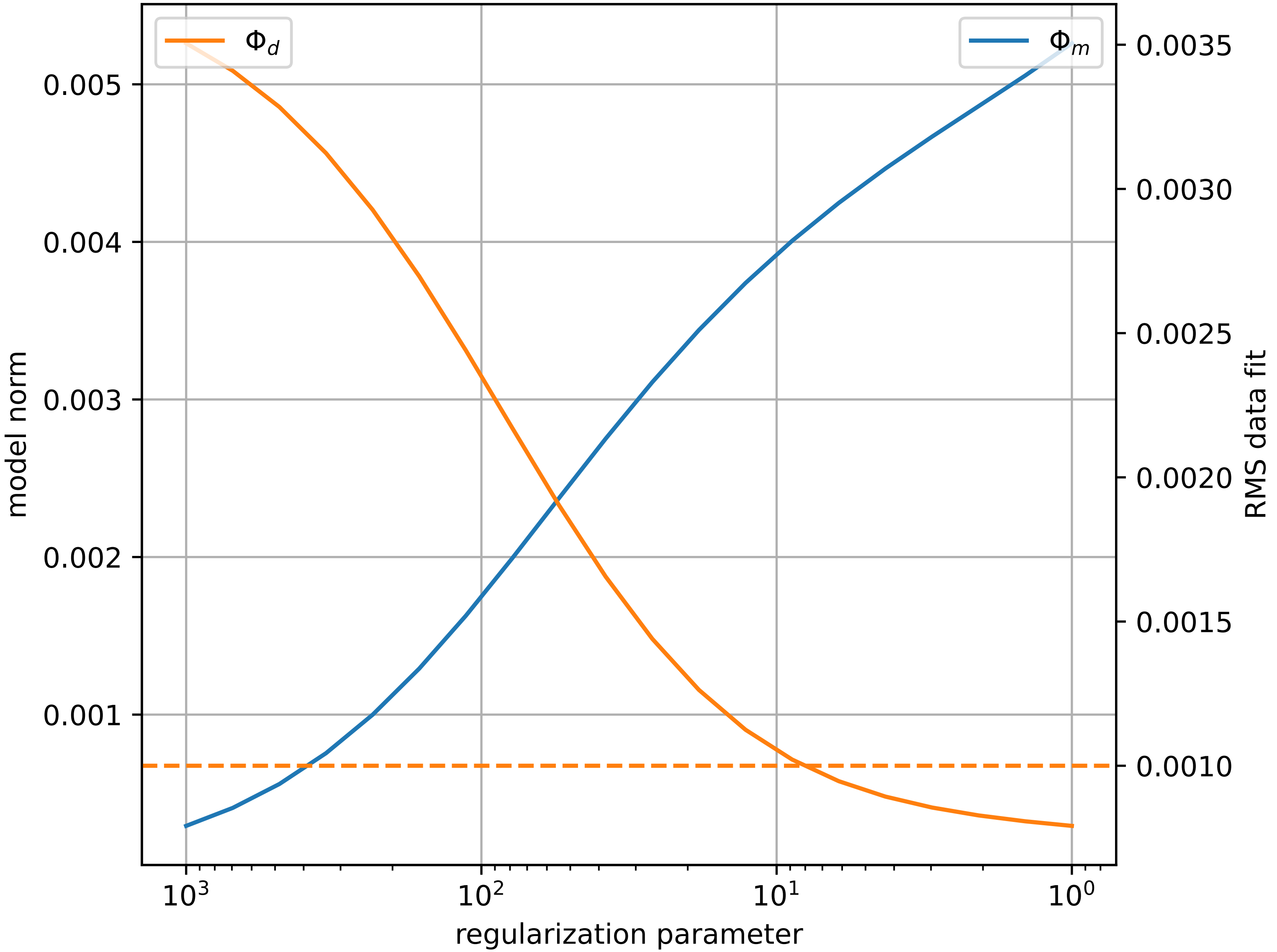
Filter factors



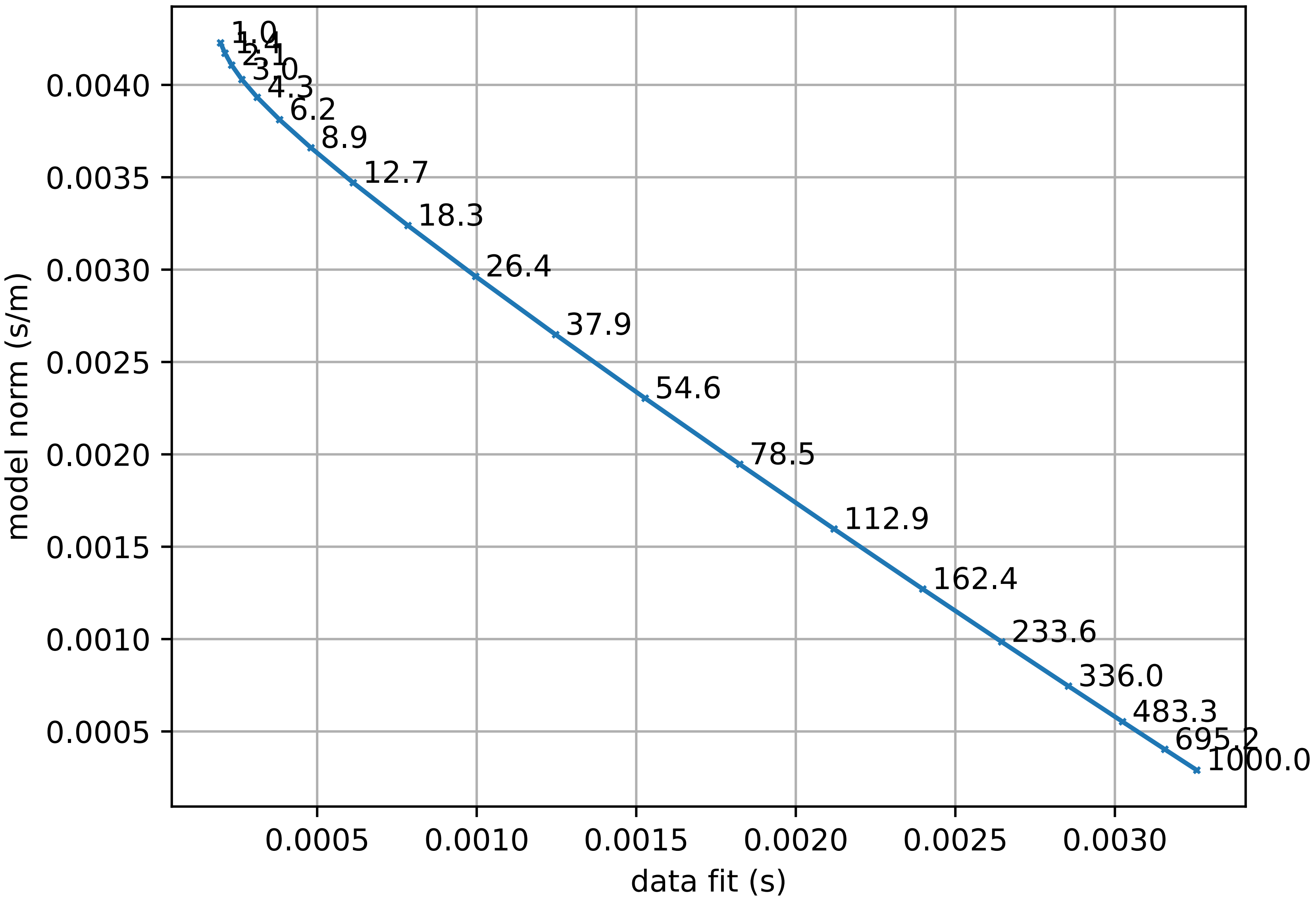
Inversion with damping



Choosing λ: Data and model norm



The L-curve



model norm (s/m)
data fit (s)
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
0.0010
0.0005
0.0005
0.0010
0.0015
0.0020
0.0025
0.0030
1.0
1.4
2.1
3.0
4.3
6.2
8.9
12.7
18.3
26.4
37.9
54.6
78.5
112.9
162.4
233.6
336.0
483.3
695.2
1000.0

Resolution of regularized inverse problems

For $c = 0$ we have $\mathbf{G}^{\dagger} = (\mathbf{G}^T\mathbf{G} + \lambda\mathbf{C}^T\mathbf{C})^{-1}\mathbf{G}^T$

$$\Rightarrow \mathbf{R}^M = \mathbf{G}^{\dagger}\mathbf{G} = (\mathbf{G}^T\mathbf{G} + \lambda\mathbf{C}^T\mathbf{C})^{-1}\mathbf{G}^T\mathbf{G}$$

approaches $\mathbf{I}$ for $\lambda \to 0$ and deviates if λ grows

Resolution for damped normal equations

$$\mathbf{R}^M = \mathbf{V} \cdot \mathrm{diag}(f_i) \cdot \mathbf{V}^T$$

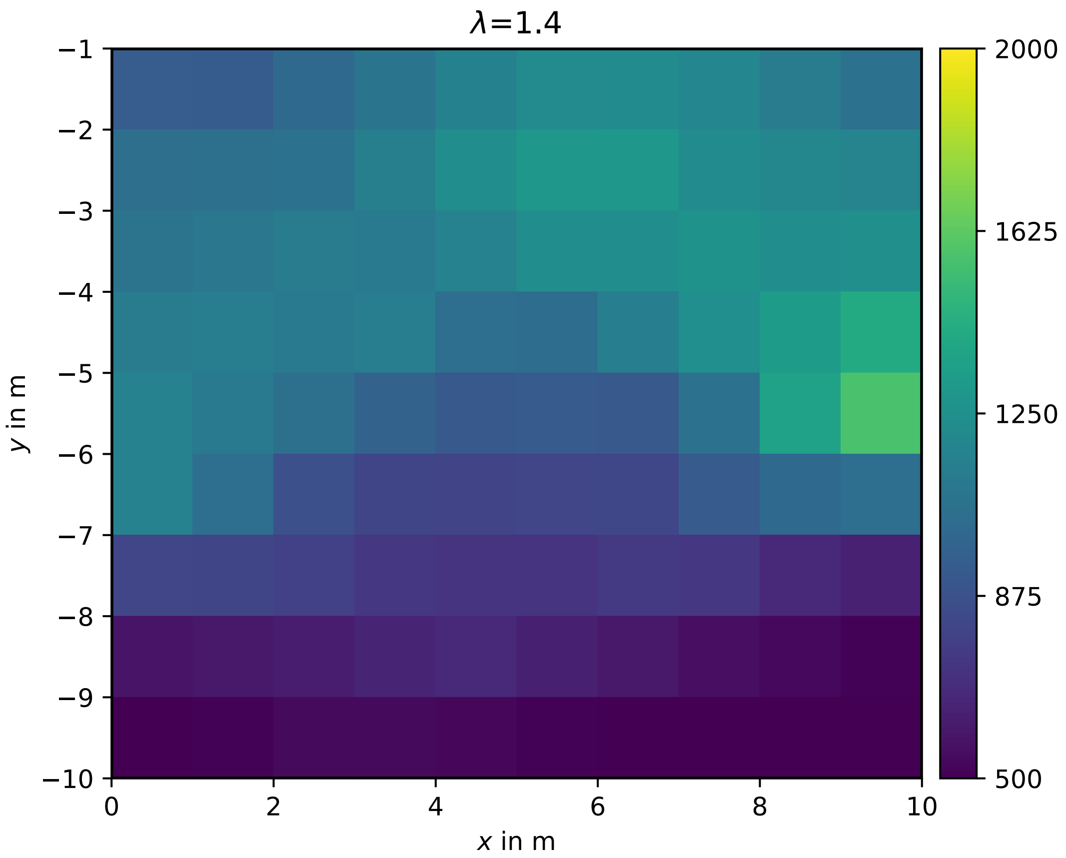
$$\mathbf{R}^D = \mathbf{U} \cdot \mathrm{diag}(f_i) \cdot \mathbf{U}^T$$

$\Rightarrow$ like for TSVD with $f_i = [1, \ldots, 1, 0, \ldots, 0]^T$

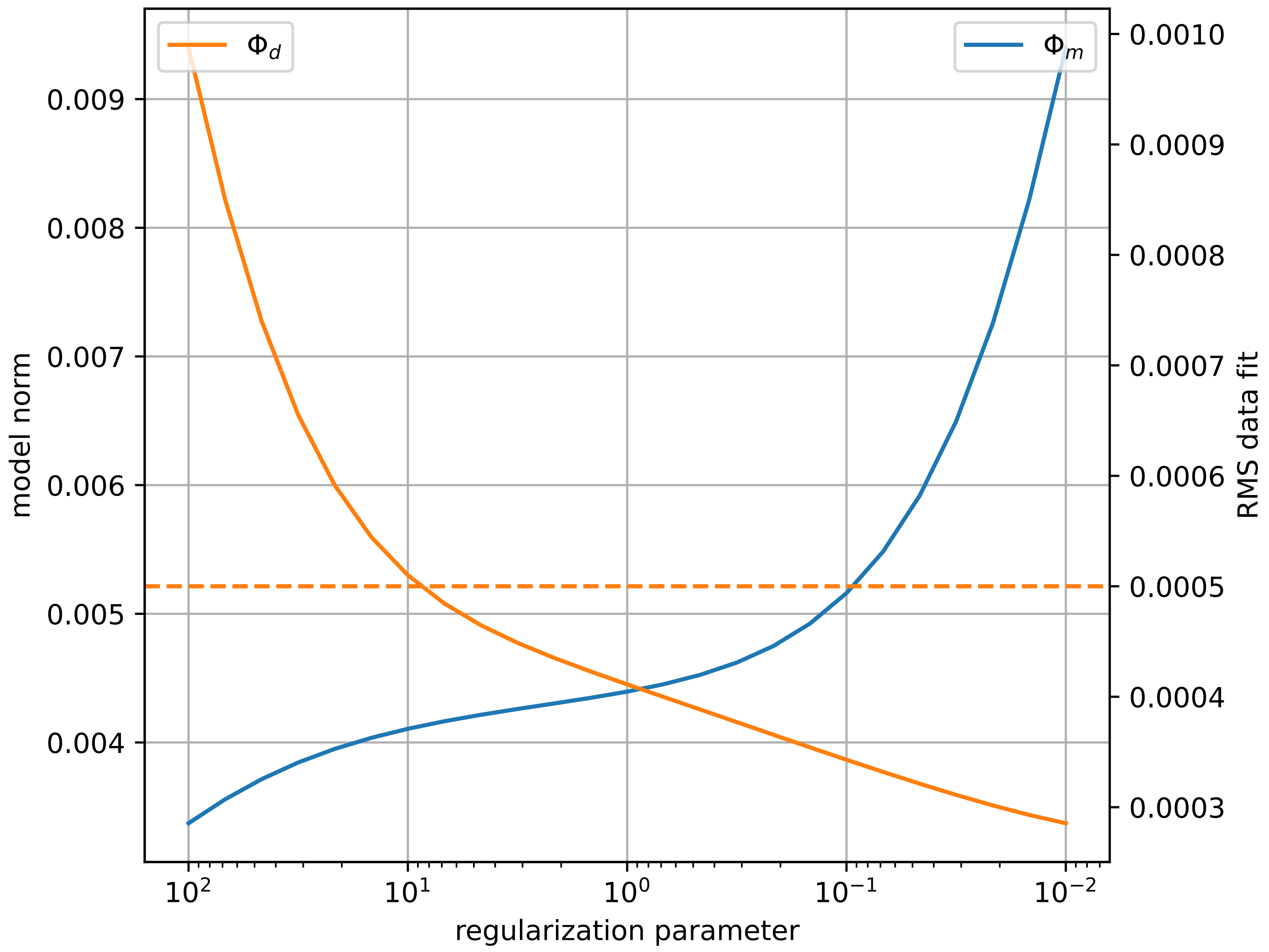
$$\mathbf{R}^M = \mathbf{V}_p \mathbf{V}_p^T \quad \text{and} \quad \mathbf{R}^D = \mathbf{U}_p \mathbf{U}_p^T$$

$$\mathbf{R}^M - \mathbf{I} = \mathbf{V}_p \mathbf{V}_p^T - \mathbf{V}\mathbf{V}^T = -\mathbf{V}_0 \mathbf{V}_0^T$$

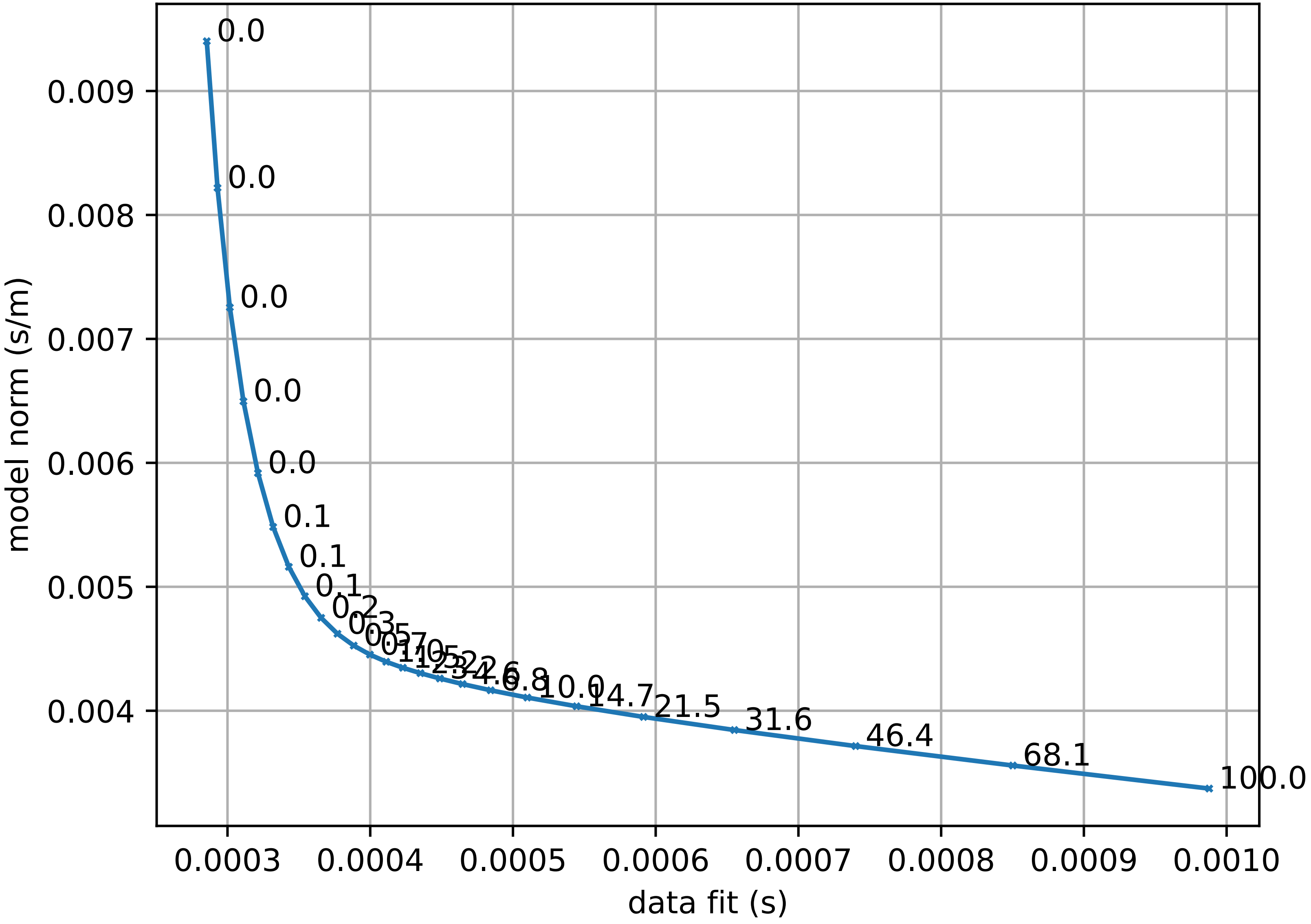
Inversion with smoothness constraints



Smoothness constraints: Data and model norm



Smoothness constraints: The L-curve

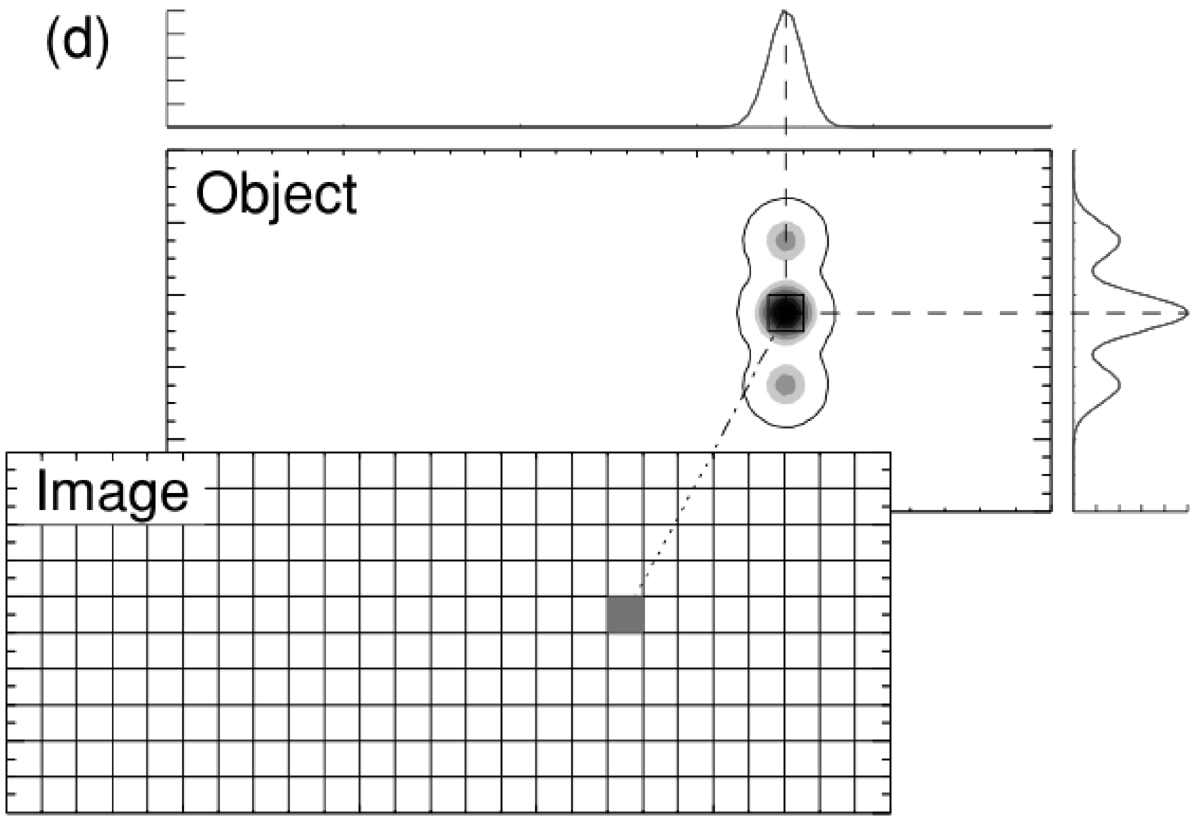


Resolution kernels (point spread functions)

$$\Rightarrow \mathbf{R}^M = \mathbf{G}^\dagger \mathbf{G} = (\mathbf{G}^T\mathbf{G} + \lambda \mathbf{C}^T\mathbf{C})^{-1}\mathbf{G}^T\mathbf{G}$$

A column represents how a cell anomaly is "spread".

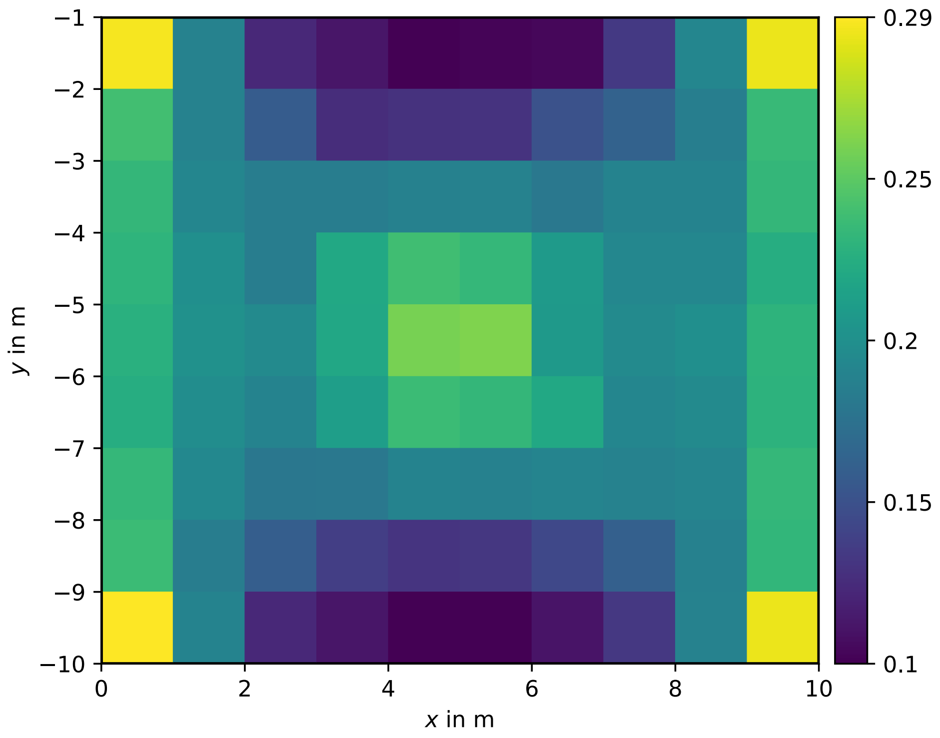
- Ideal imaging
- Contrast deficiency
- Geometrical distortion
- Side lobes



Model resolution radius

The diagonal of the resolution matrix

states how a single model cell can be retrieved



How big needs a sphere to be to integrate a value of 1 (perfect resolution)?

Resolution radius

$$r = \frac{r_i}{\sqrt{\mathbf{R}^M_{ii}}} = \frac{\sqrt{A_i/\pi}}{\sqrt{\mathbf{R}^M_{ii}}}$$